

CPSC 8440: Generative AI: Technical and Human Perspectives

Module 1.2: Introduction to NLP

Natural Language Processing

- NLP = building **computer programs** to analyze, understand and generate **human language - either spoken or written** (informal)
- NLP is an interdisciplinary field

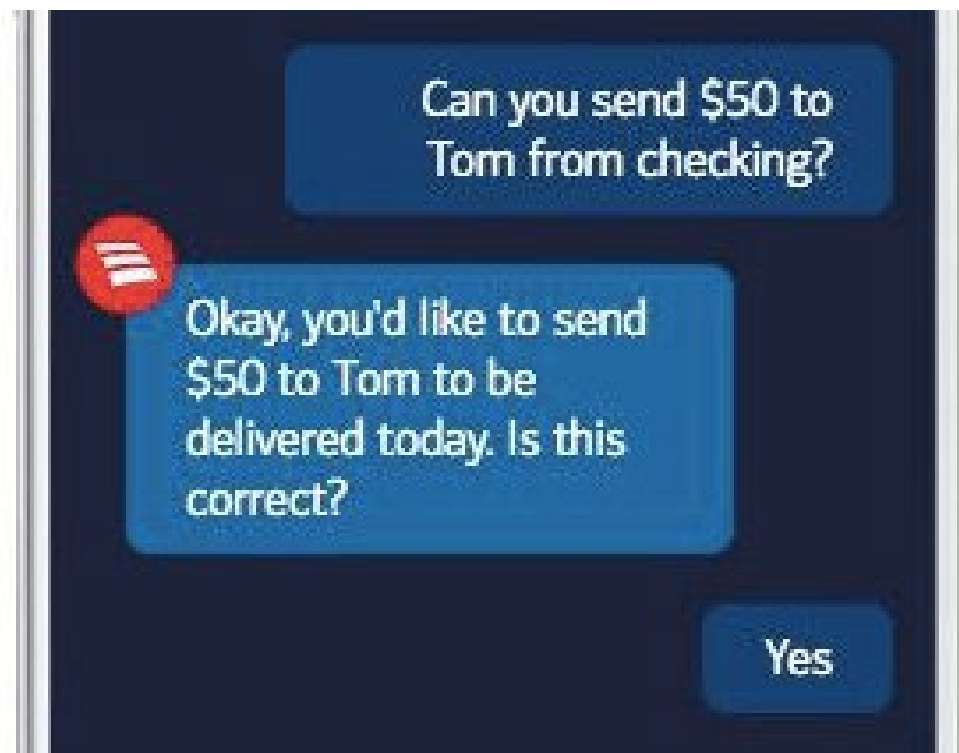


Natural Language Processing

- NLP = building **computer programs** to analyze, understand and generate **human language - either spoken or written** (informal)

Communication with humans (ex.
personal assistants, customer service)

Access the wealth of information about
the world — crucial for AI systems



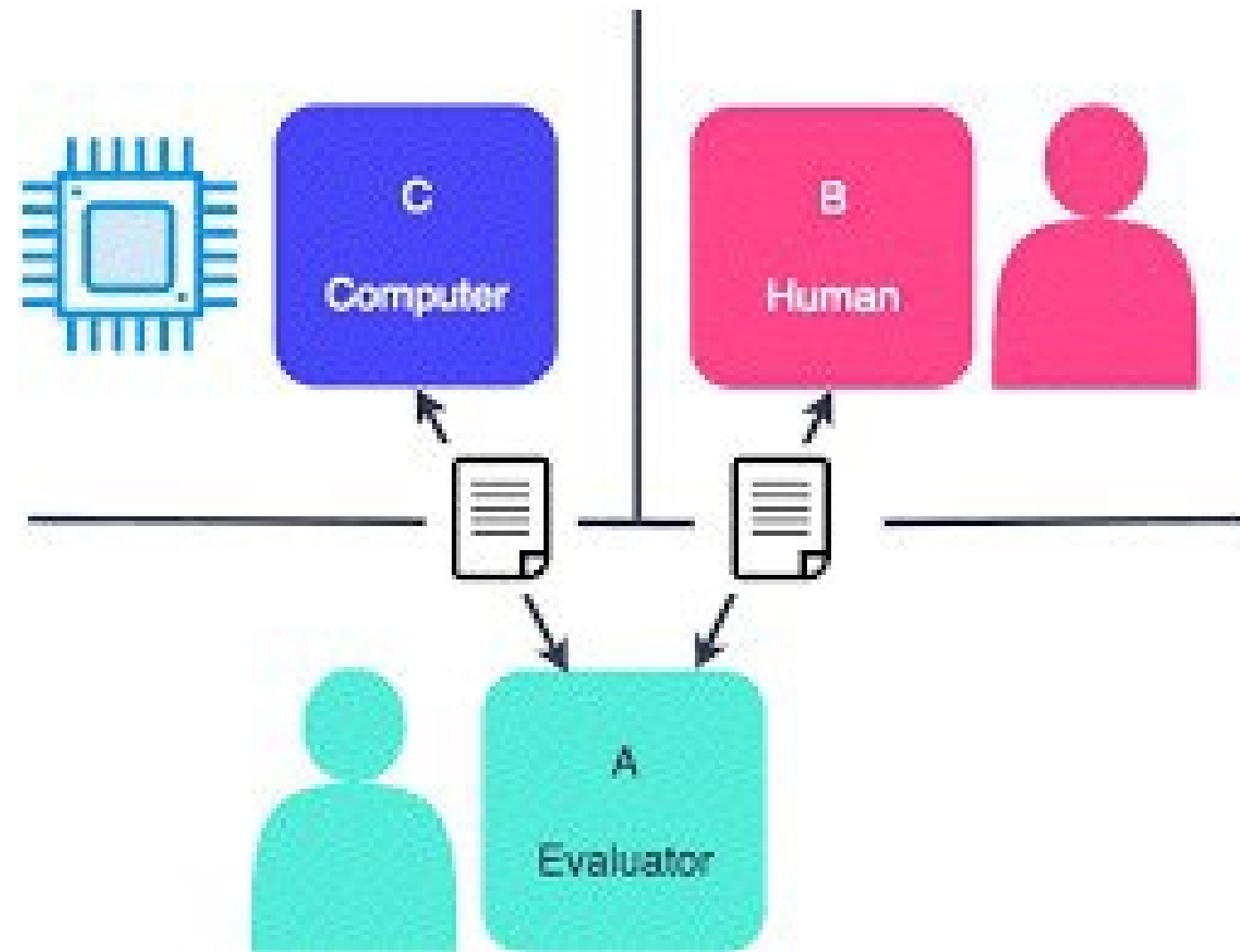
Banking assistant

ONLINE



OFFLINE

Turing Test

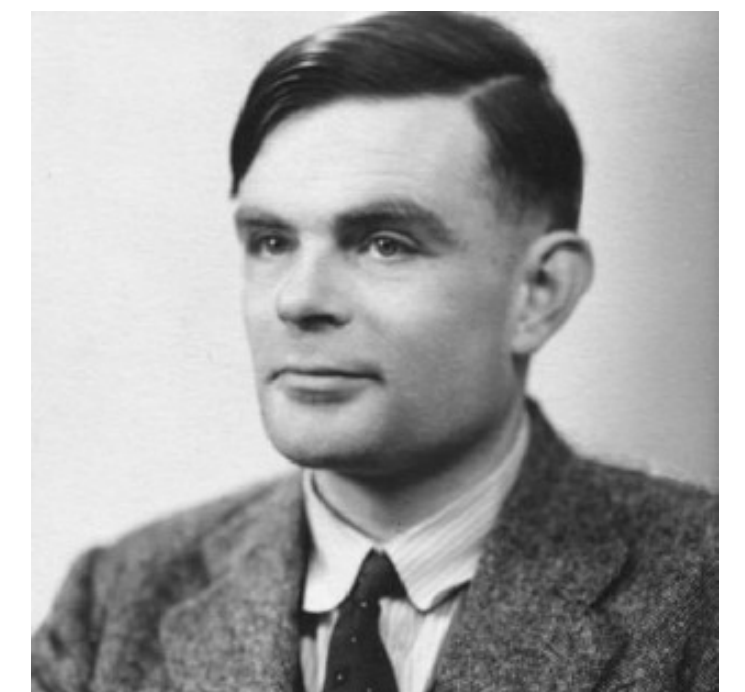


A. M. Turing (1950) *Computing Machinery and Intelligence*. *Mind* 49: 433-460.

COMPUTING MACHINERY AND INTELLIGENCE

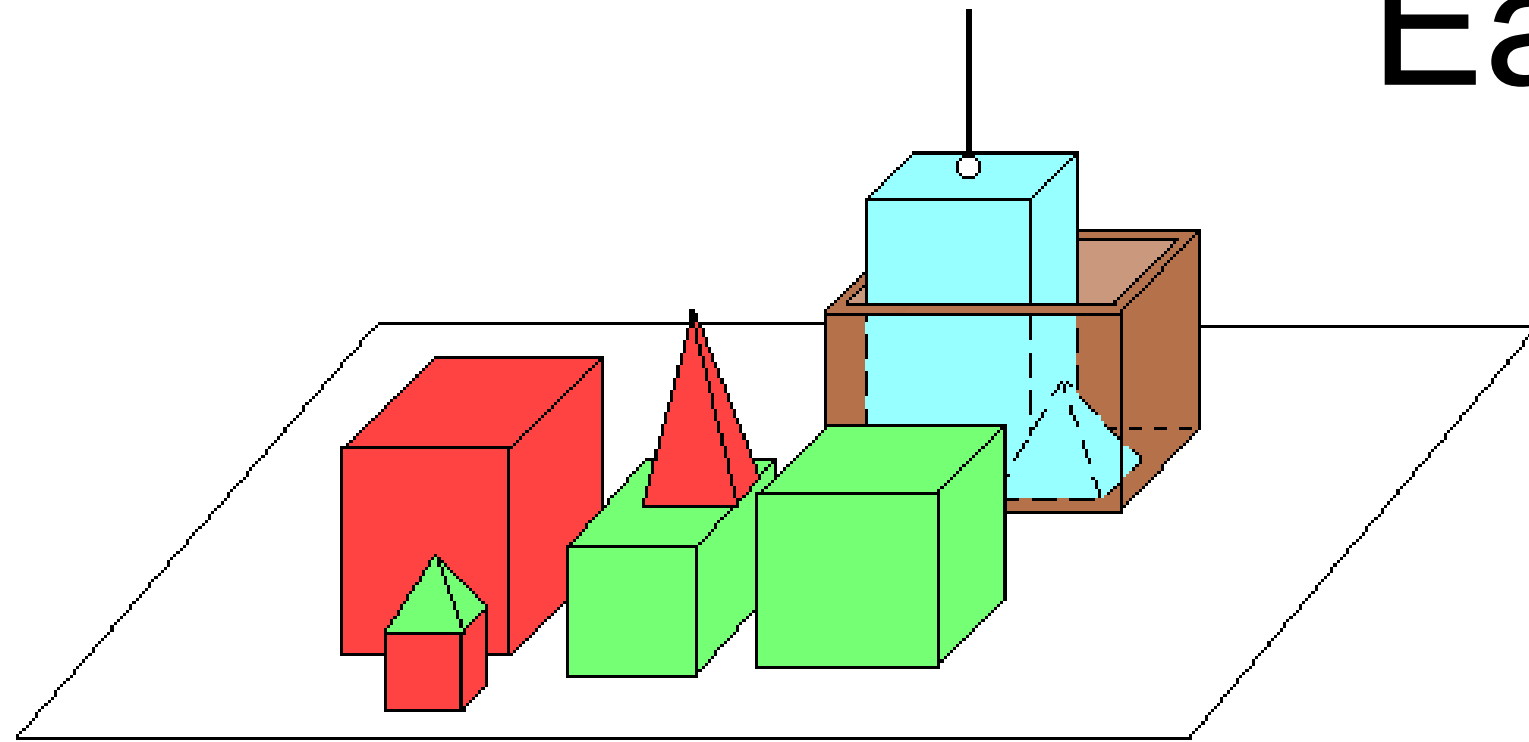
By A. M. Turing

1. The Imitation Game



Ability to understand and generate language ~ intelligence

Early NLP



SHRDLU,
1968

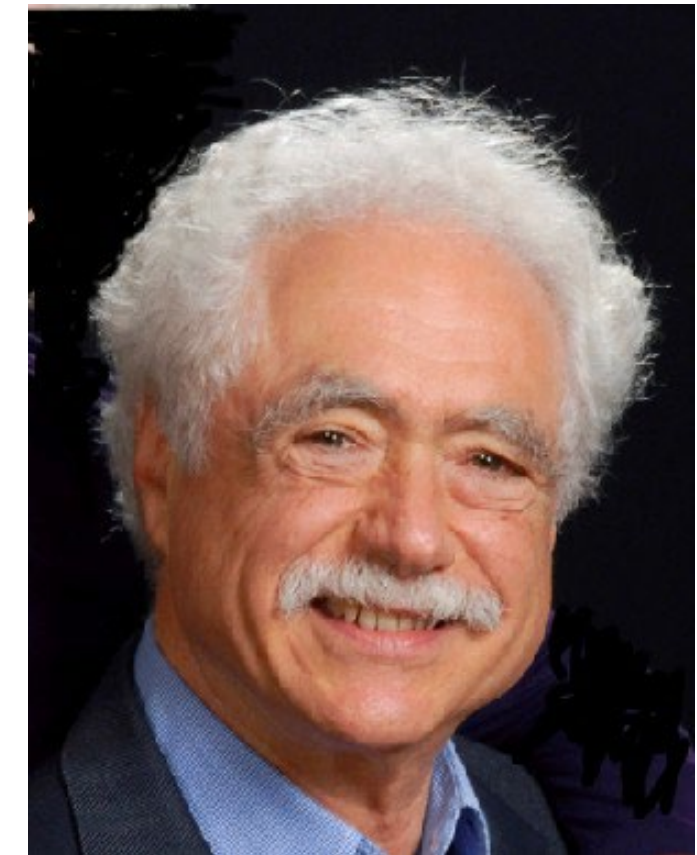
> How many red
blocks are there?

- THREE OF THEM

> Pick up the red
block on top of a
green one

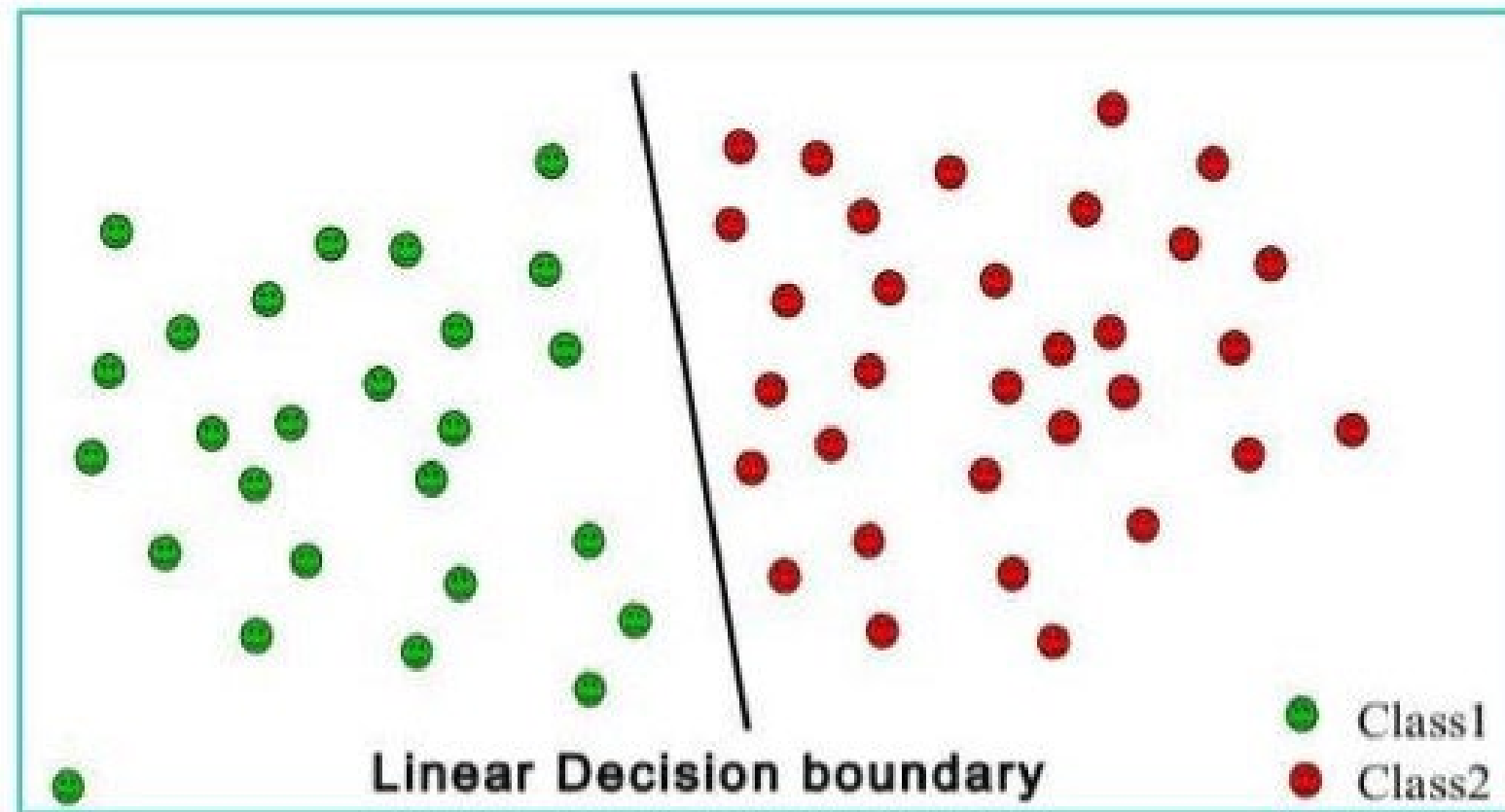
OK.

- Rule-based, requiring extensive programming
- Limited domain



Terry Winograd

Statistical learning



- Use of machine learning techniques in NLP
- Increase in computational capabilities
- Availability of electronic corpora

The era of deep learning

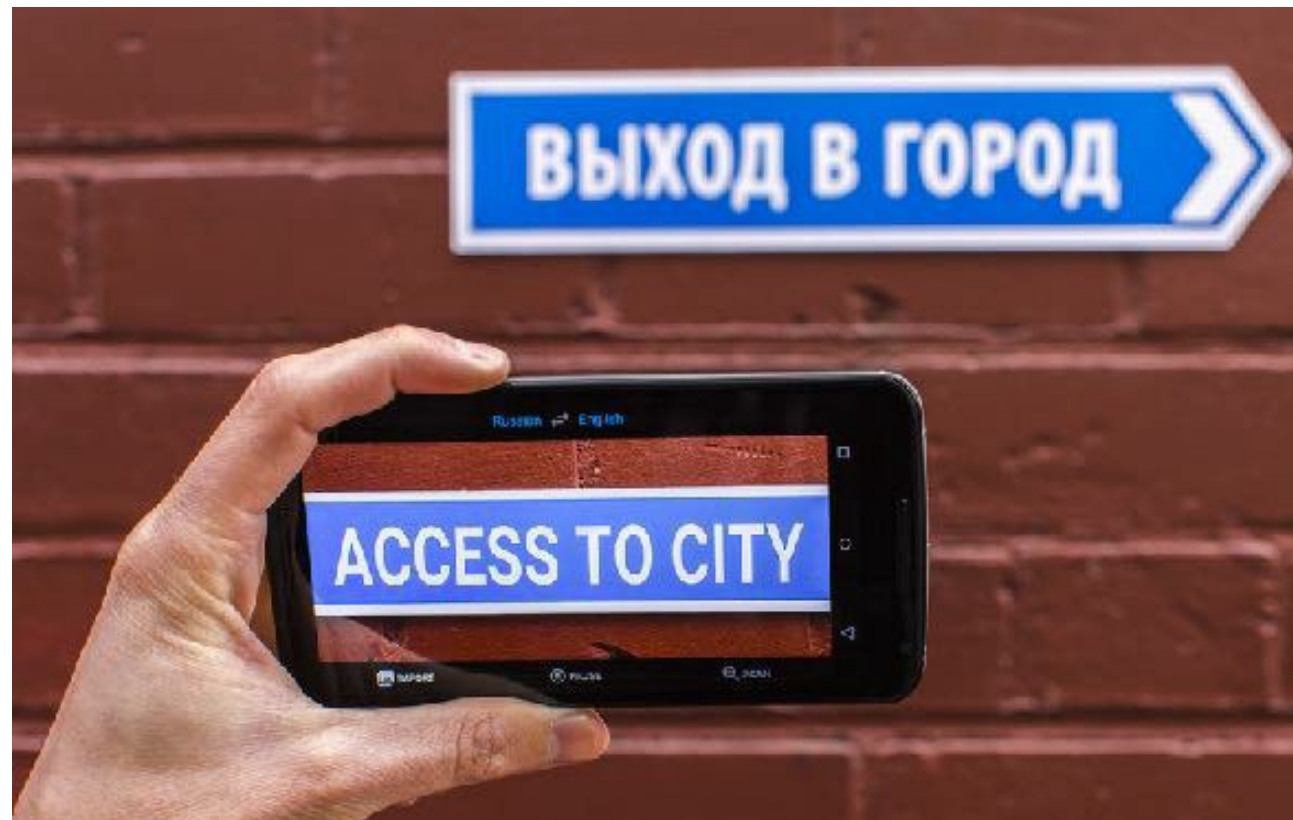
- Significant advances in core NLP technologies
- **Essential ingredient:** large-scale supervision, lots of compute
- Reduced manual effort - less/zero **feature engineering**



GPU



TPU



36M sentence pairs

Russian: Машинный перевод - это круто!

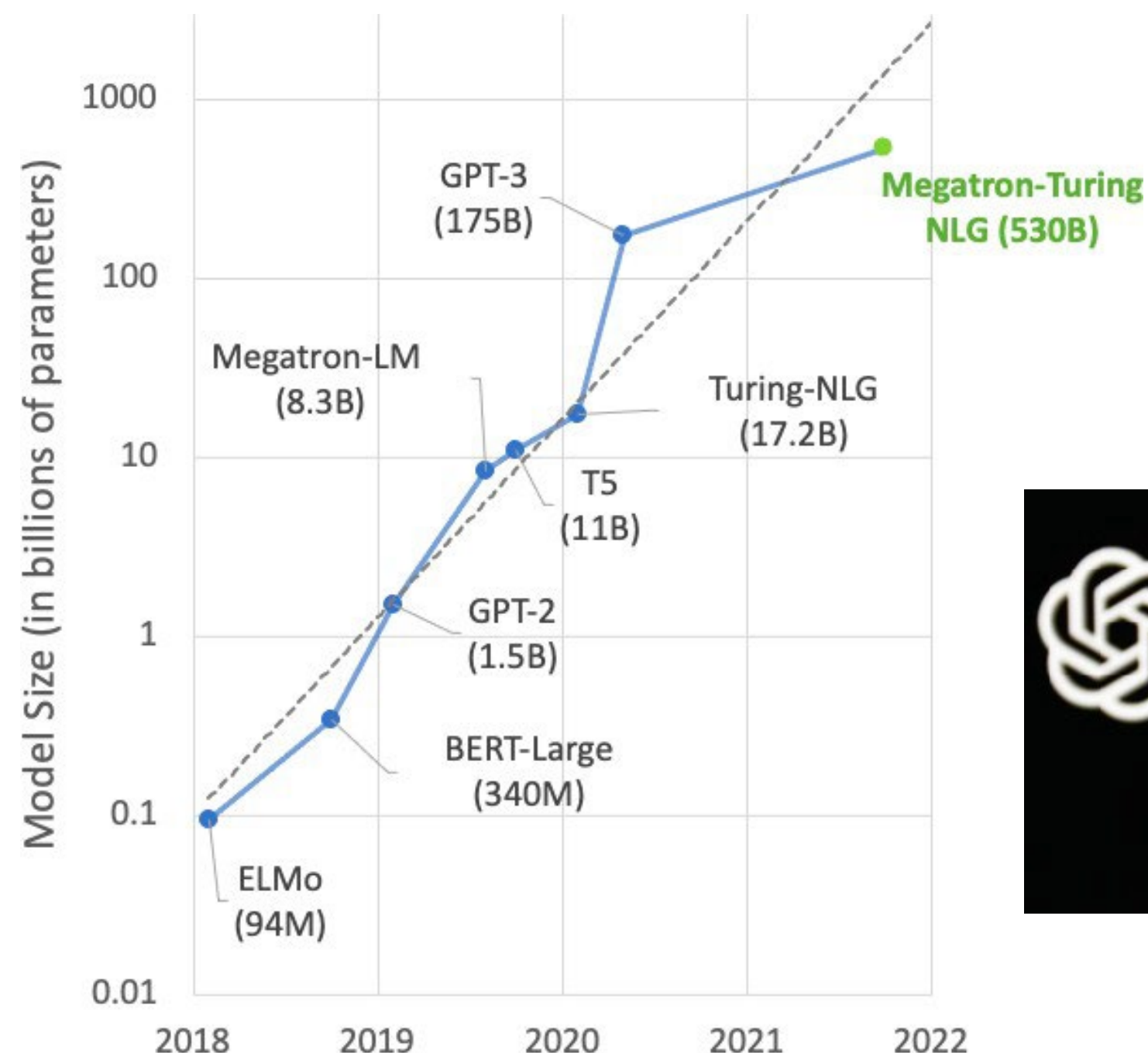


English: Machine translation is cool!

The era of pretraining / LLMs



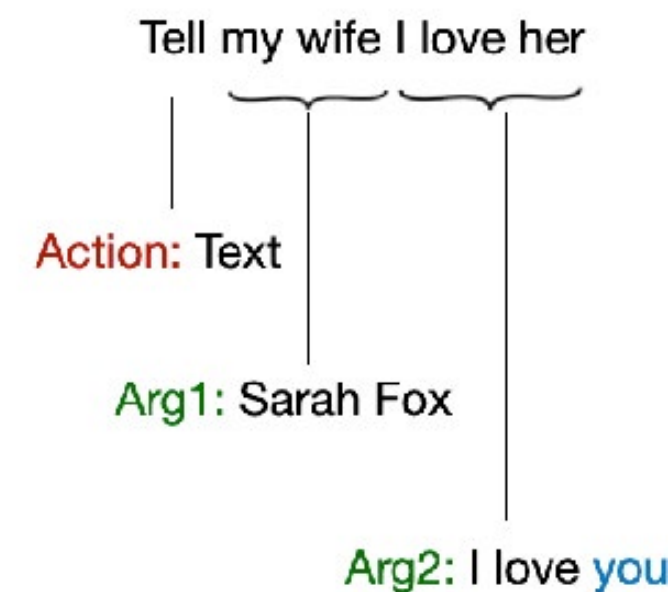
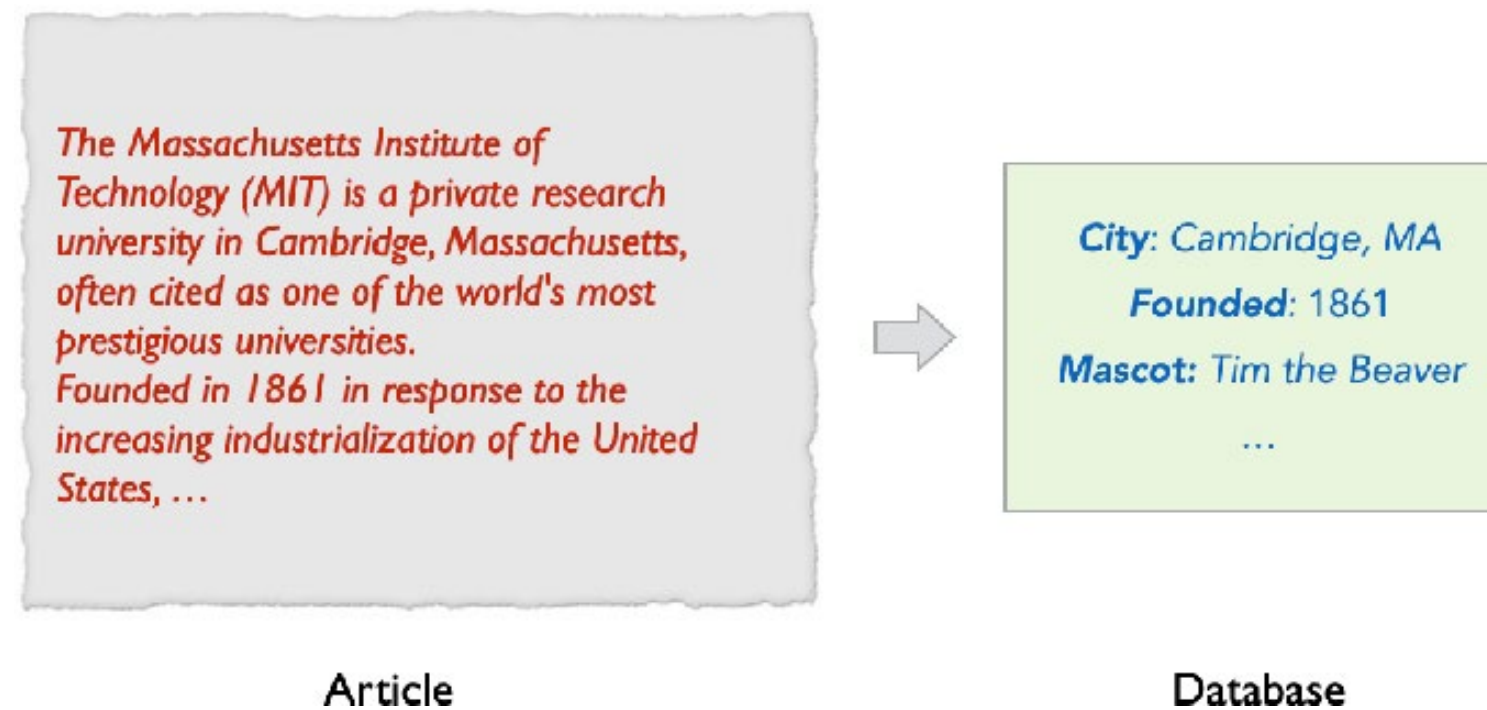
BERT, ELMo, ERNIE...



- Leverages a lot of unlabeled text
- Model size increased by $10^3 - 10^5$ x in parameters

NLP subfields

- Machine translation
- Question answering
- Information extraction
- Text summarization
- Dialogue
- Semantic parsing
- Syntactic parsing
- Language grounding
-



NLP topics we will cover in this class

We will discuss (very few) fundamental, important concepts before learning LLMs

- Word embeddings
- Traditional language models
- Transformers
- **Large language models**